

Transregio SFB-TR40 – Test Case 1

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Test Case 1

Single Element

Combustion Chamber – GCH₄/GOX

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1 General presentation

The demanding issues in terms of high operational and handling costs of cryogenic and storable propellants increased the attention for oxygen/methane in the development of future launch vehicles.

Although liquid oxygen/methane propellant combination is an attractive option, only a limited amount of experimental data is available at relevant combustion chamber conditions and a critical gap is present in the knowledge of detailed heat transfer characteristics and injector technology.

In the context of the national research program Transregio SFB/TR-40, two multi-injector combustion chambers have been designed for gaseous oxygen (GOX) and gaseous methane (GCH₄), respectively focusing in TP K1 on high pressure (up to 100 bar) heat transfer and film cooling behaviour and in TP D9 on low pressure (up to 40 bar) application for optical measurement techniques. One of the key aspects of these projects is to improve the knowledge on heat transfer processes, in particular the injector–injector and injector–wall interactions, and cooling methods at representative rocket engine conditions. As a first step of the experimental effort a single element combustion chamber has been manufactured and tested, featuring similar characteristics as the aforementioned test hardware like an identical injector design and an equal chamber contraction ratio.

The principal aim of the test campaign is to experimentally document the detailed wall heat flux characteristic of a gaseous oxygen/gaseous methane shear coaxial single-element injector over a range of pressures and mixture ratios. This set of data is a benchmark, suited for the validation purposes of numerical tools. A first effort in this direction has already been reported in [1]. In order to provide a well-proven data set as a test case, for validation of models and numerical methods to the Transregio and the combustion modelling community beyond, a great effort is put into cross-checking all individual measurements and predicted data.

The details for both the test facility geometry and operating conditions required for the numerical simulations are listed below.

2 Geometry

2.1 Combustion chamber

In this section a description of the instrumented single element rocket chamber, injector geometry and flow conditions used for the wall heat flux characterization, is presented.

The combustion chamber has an inner square cross section of 12 mm x12 mm. It consists of two chamber segments of 174 mm and 145 mm length and a nozzle segment of 20 mm length. The nozzle differs from a usual configuration due to its throat with a rectangular cross section of 4.8 mm x 12 mm, which results in a contraction ratio of 2,5 and Mach number in the chamber of 0,24.

The entire assembly is capacitively cooled in order to keep the design flexible and more accessible for temperature measurements. The complete chamber geometry is depicted in Figure 1 and the main characteristic dimensions are summarized in Table 1.

Combustion chamber dimensions		
Chamber length **	[mm]	303
Chamber internal width	[mm]	12
Chamber internal height	[mm]	12
Chamber external width	[mm]	85
Chamber external height	[mm]	50
Throat width	[mm]	12
Throat height	[mm]	4,8
Contraction ratio	[-]	2,5
Characteristic length (L^*)	[mm]	749
**from faceplate to throat		

Table 1 Combustion Chamber Characteristic Dimensions

More detailed drawings about the chamber geometry can be found in Annex 1.

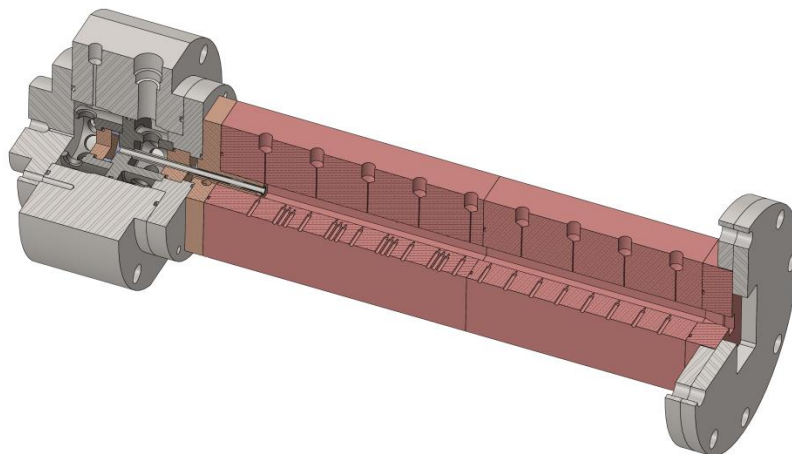


Figure 1 Combustion Chamber Geometry

The material used for the chamber segments and the nozzle segment is oxygen-free copper (Cu-HCP). The main characteristics of this material are reported in Table 2.

Material Property		Cu-HCP
Density	[kg/m ³]	8940
Thermal expansion coefficient	[°C ⁻¹]	1,70E-05
Thermal conductivity	[W/mK]	385
Tensile yield stress	[MPa]	200
Tensile ultimate stress	[MPa]	310
Young modulus	[GPa]	112,5

Table 2: Material Properties

The ignition of the chamber is achieved by a torch igniter using GCH₄/GOX, mounted to the side wall of the combustion chamber at 130 mm from the faceplate, perpendicular to the main flow stream direction. The position of the igniter has been modified after further investigation on the influence on the flame behavior as reported in [2].

The mass flow rates in the combustion chamber (GCH₄, GOX) are set by sonic orifices in the feed lines and the upstream pressure. The orifices in the feed lines to the main injector have been manufactured with appropriate diameters and calibrated with nitrogen using a Coriolis flow meter prior to the test campaign. The actual mass flows are calculated from the recorded pressure, temperature signals and the orifice calibration data after the test, assuming sonic flow condition in the orifices.

2.2 Injector

For the current test case, a single shear coaxial injector element is integrated in the combustion chamber, where the core of gaseous oxygen is surrounded by high speed gaseous methane. To ensure homogeneous injection conditions, in terms of temperature and pressure, two porous plates are placed in the oxidizer and fuel manifolds respectively as shown in Figure 2.

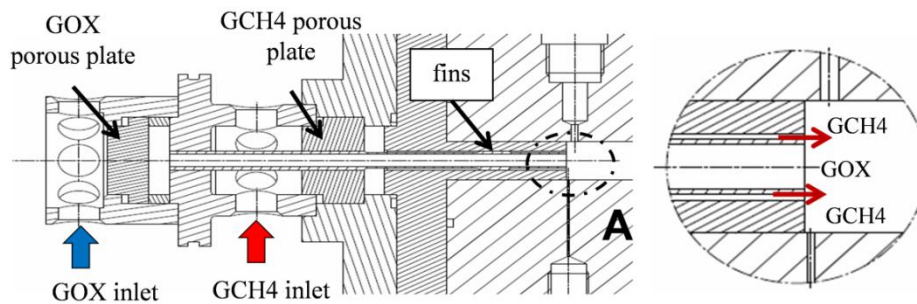


Figure 2: Shear Coaxial Injector

For simplicity, the GOX post is configured flush with respect to the injection face (no recess).

In order to center the injector element in the faceplate, it has been equipped with four equally-spaced fins, 14 mm upstream the injection point. For the current test series, the fins are positioned with an angle of 45° to the combustion chamber center plane.

Main injector dimensions are reported in Table 3.

Injector dimensions		
Type	[-]	shear coaxial
GOX internal diameter	[mm]	4
GCH ₄ internal diameter	[mm]	5
GCH ₄ external diameter	[mm]	6
GOX post wall thickness	[mm]	0,5
GOX recess	[mm]	0
GOX tapering	[deg]	0

Table 3: Injector Dimensions

Further information about the hardware and the combustion chamber set-up can be found in [3].

3 Sensors

3.1 Feed line measuring positions

In the current section the main measuring sensors located in the feed lines are described.

Table 4 reports the different sensors, their nomenclature and the measuring positions.

Sensor	Name	Position	Type	Precision
Thermocouple	TO2	feed line oxygen\prior to orifice	K	$\pm 1,5$ K
Thermocouple	TO3	manifold oxygen\prior porous plate	T	$\pm 0,5$ K
Thermocouple	TM2	feed line methane\prior to the orifice	K	$\pm 1,5$ K
Thermocouple	TM3	manifold methane\prior porous plate	T	$\pm 0,5$ K
Pressure transducer	PO1	feed line oxygen	Sensor Technics	350 mbar
Pressure transducer	PO2	feed line oxygen\prior the orifice	Sensor Technics	100 mbar
Pressure transducer	PO3	manifold oxygen\prior porous plate	Sensor Technics	50 mbar
Pressure transducer	PO4	manifold oxygen\prior porous plate	Sensor Technics	50 mbar
Pressure transducer	PM1	feed line methane	WIKA A 10	250 mbar
Pressure transducer	PM2	feed line methane\prior the orifice	WIKA A 10	100 mbar
Pressure transducer	PM3	manifold methane\prior porous plate	WIKA A 10	60 mbar
Pressure transducer	PN1	feed line nitrogen	WIKA A 10	250 mbar
Pressure transducer	PF1	purge film applicator	WIKA A 10	40 mbar
Pressure transducer	PMZ	feed line methane ignitor\prior the orifice	WIKA A 10	100 mbar
Pressure transducer	POZ	feed line oxygen ignitor\prior the orifice	Sensor Technics	100 mbar
Pressure transducer	PCZ	feed line oxygen ignitor\prior the orifice	WIKA A 10	60 mbar

Table 4: Feed Line Measuring Positions

Figure 3 shows a schematic of the feed lines of the test facility, for major clarity.

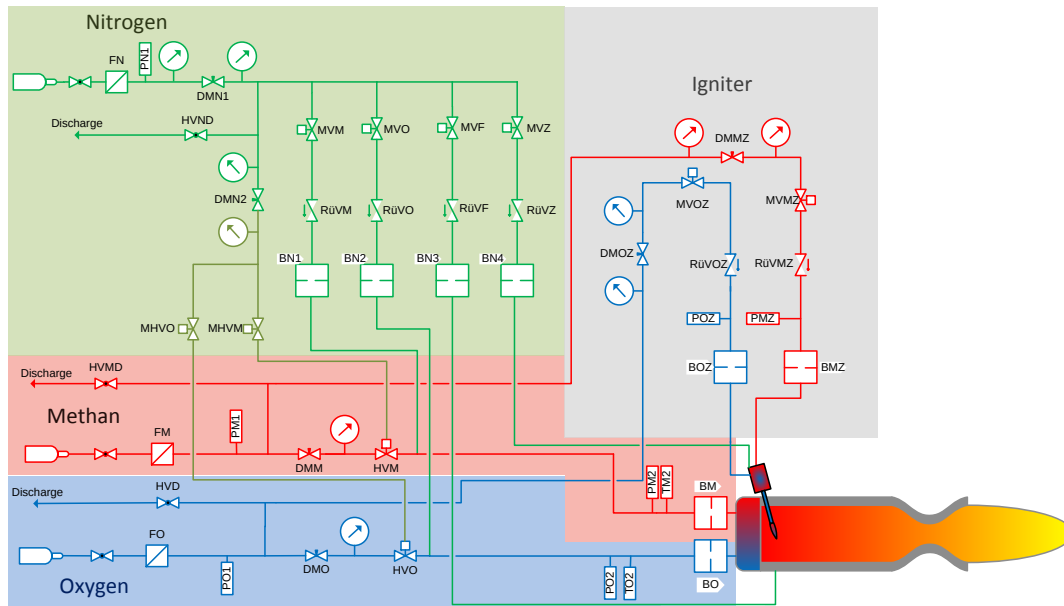


Figure 3: Test Facility Feed Line Schematic

The pressure sensors are individually calibrated and only the 0.1% error on the measuring sensor range, due to the non-repeatability, has to be considered for the sensor accuracy. Particular attention is made in the calibration of the pressure transducers PO2 and PM2, used for the mass flow rate calculation, where the complete measuring chain is calibrated.

3.2 Combustion chamber measuring positions

In the present paragraph the different measuring positions and sensors, with which the combustion chamber is instrumented, are shown.

To characterize the injection conditions, thermocouples and pressure transducers are installed in the chamber manifolds, upstream the porous plates. The square region of the combustion chamber is instead equipped with a series thermocouples and wall pressure measurements.

Equally spaced pressure transducers, on the side wall, provide a well resolved measurement of the evolution of the wall pressure distribution $p(x)$ along the chamber axis for a better understanding of the complex heat release and combustion processes. A schematic of the combustion chamber and the associated sensor locations is given in Figure 4. In the figure are reported also the coordinate system useful for the identification of the different sensors positions.

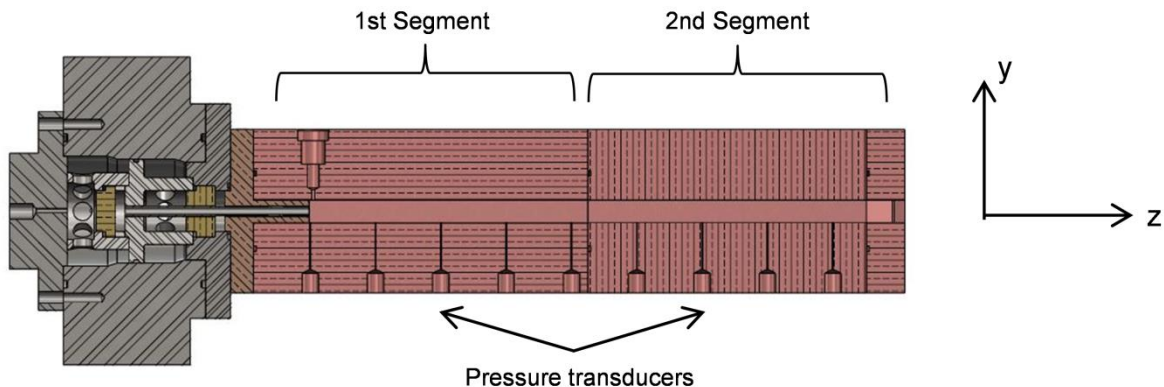


Figure 4: Pressure Measurement Positions

To determine the temperature field, thermocouples are applied in the chamber walls in three different ways: type T thermocouples of 0.5 mm diameter located within the chamber wall with 1 mm, 2 mm and 3 mm distance to the hot wall, additionally two coaxial Type T thermocouples (Medtherm), flush mounted with the hot wall, and five Type K surface thermocouples attached to the external surface. The Type T thermocouples are mounted with a regular pattern of 17 mm in the upper surface of the first and second segment, along the center plane of the combustion chamber. The Medtherm thermocouples are instead press-fitted into the chamber wall, in corresponding axial positions in the lower surface of the first segment. The thermocouple location pattern is shown in Figure 5. Furthermore four clusters of three Type T thermocouples with different distance from the hot-wall are placed in the upper side of the first segment, in order to characterize the heat flux within the chamber wall

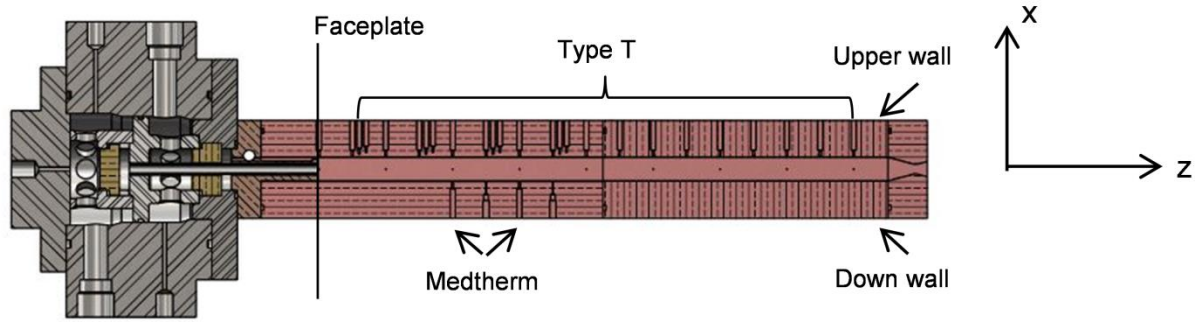


Figure 5: Temperature Measurement Positions

The thermocouples are from one batch and have a global precision of ± 0.5 K. To improve the results coming from the transient measurement of the thermocouples efforts are initiated. In a preparatory test campaign, the thermocouples have been individually dynamically classified in terms of response time and absolute values. This allows the determination of an optimal thermocouple arrangement, particularly for the thermocouple clusters used for the heat flux reconstruction in the combustion chamber walls.

Detailed information about the measuring sensors and the complete list of sensors is reported in Annex 2.

4 Test operating condition

The operating point chosen for this test case is a nominal pressure of 20 bar and mixture ratio 2.6.

The use of a heat sink hardware limits the duration of the test. The hot run is set to 3 s, in order to reach the required stable operation needed for the thermal load measurements, and preserve the integrity of the hardware. A typical pressure and temperature signal is shown in Figure 6.

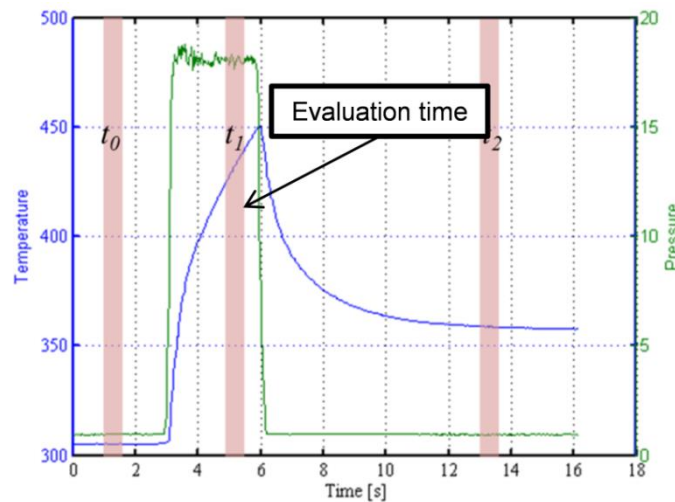


Figure 6: Typical Pressure/Temperature Reading

Due to the transient nature of the problem, a time interval representative for the hot run (t_1) at 2/3 of the burning time is established (see Figure 6). This time interval will be referred as “Evaluation time”. The operating conditions defined in Table 5 are referred to the afore mentioned time interval and averaged therefore over a hot run time of 0.5 s.

Testname	Burning_time	Evaluation_time	OF	Mdot_Ox	Mdot_CH4	Mdot_tot
[-]	[s]	[s]	[-]	[kg/s]	[kg/s]	[kg/s]
TFC-R-20-26-30-10	3,02	2,01	2,62	0,045	0,017	0,062

Table 5: Operating conditions

The reported “Burning time” is evaluated as the time interval between the first time step at which the recorded value for the pressure in the combustion chamber is bigger than 80 % of the ambient pressure and the first time step at which its value is less than 80%. The “Evaluation time” is considered as the time from the start of the hot run in which the evaluation time interval is centered.

The orifices used to set the entry mass flow rate as well as the pressure sensor are individually calibrated. The mass flow rate values, calculated by means of the pressure and temperature upstream, present a maximum absolute error of 0.98 g/s for methane and 1.67 g/s for oxygen.

The entry conditions for GOX and GCH4 are:

PO3	TO3	PM3	TM3
[bar]	[K]	[bar]	[K]
26,4	278	24,4	269

Table 6: Propellant entry conditions

The values reported in Table 6 are acquired from sensors positioned upstream the porous plates (see Section 2.1). For these porous plates a pressure drop of about the 9% of the entry pressure for the oxygen side and of the 11% for the methane side has to be taken into account. The temperature instead can be assumed constant.

5 Boundary conditions

In order to define proper boundary conditions, the wall temperatures are reconstructed by means of the use of the in-house tool Thermtest [4], based on numerical correlations. The temperature profile along the chamber axis is discretized every 10 mm and given in two different points of the combustion chamber cross section: middle upper plane (T_{up}), where the line of the thermocouples has also been installed in the chamber wall, and in the middle side plane (T_{side}), as shown in Figure 7. This choice is done in order to take into account the different thickness of the combustion chamber walls and the effect due to the presence of a corner. The corresponding values are reported in Annex 3: Wall and Thermocouple Temperatures.

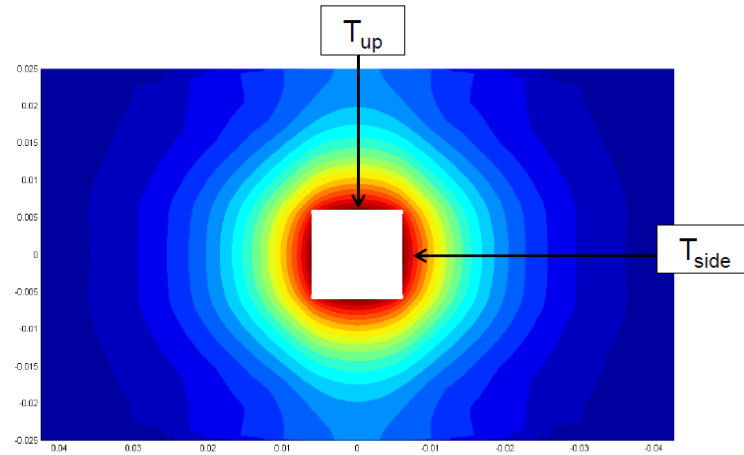


Figure 7: Wall Temperatures Positions

Furthermore the values for the thermocouples reading at the evaluation time (see Figure 6) are also reported in Annex 3: **Wall and Thermocouple Temperatures**. The discretization in this case is of 17 mm for the up wall while only 4 measuring positions are present in the down wall. No measuring positions are foreseen for the side wall, like already described in Section 3.2.

Requested results

The results of the numerical computations must be presented in such a way that they can be compared with experimental data. For this reason, the following information are required:

- Axial profile of the wall combustion chamber pressure
- Heat flux profiles along the chamber walls for the 2 prescribed positions (T_{up} , T_{side}).

Additionally, in order to be able to compare the different reaction schemes, the following results shall be provided:

- Iso-plots of temperatures and species
- Axial evolution of the main products (mass fraction)
- OH radical concentration along the chamber axis.

6 Annex 1: Chamber Geometry

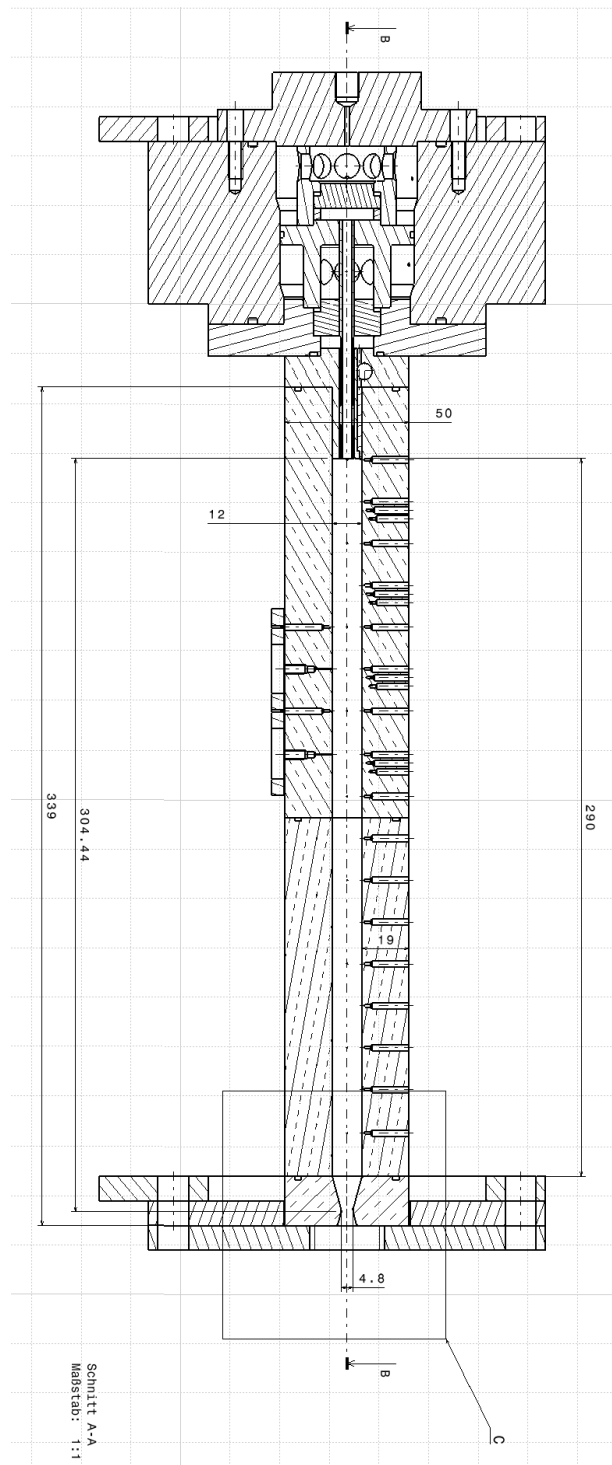


Figure 8: Combustion Chamber Cross Section A-A

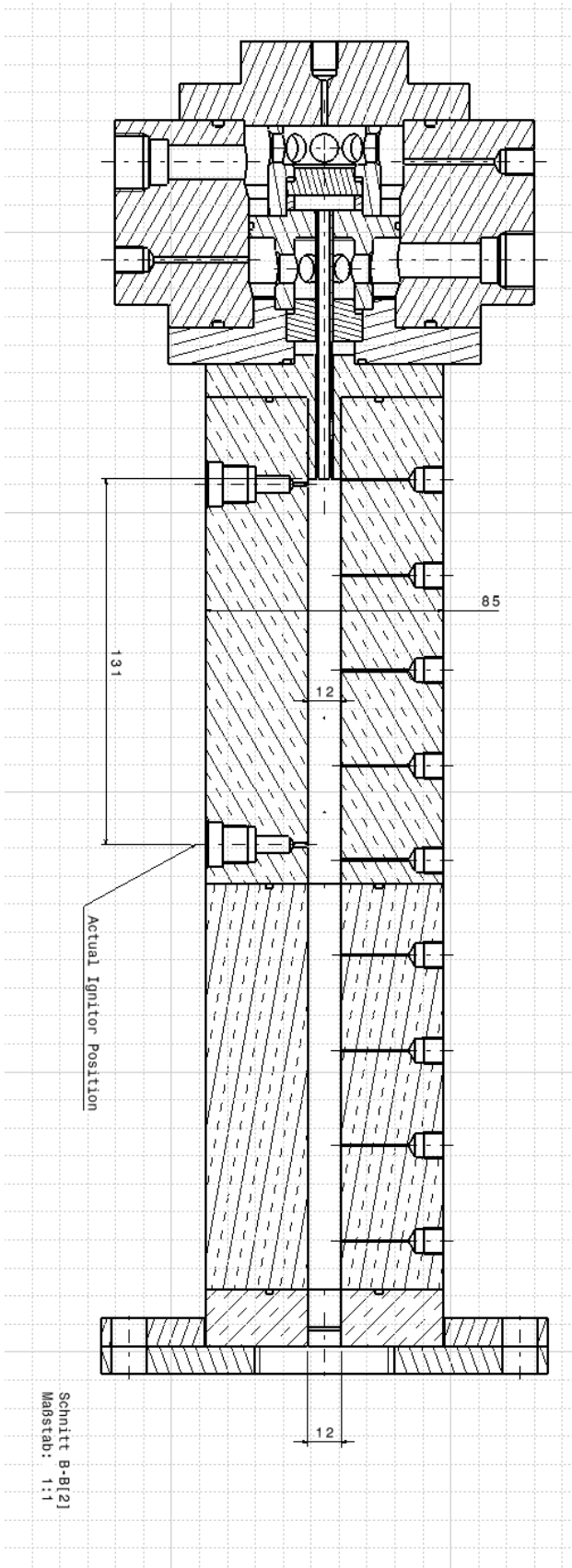


Figure 9: Combustion Chamber Cross Section B-B

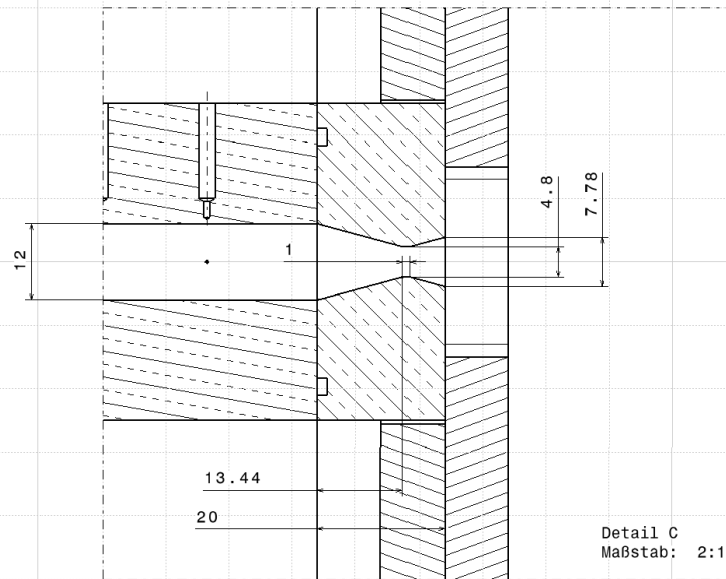


Figure 10: Nozzle – Detail C

7 Annex 2: Measurement Positions in the Combustion Chamber

Sensor	Name		Position		Type	Precision
			D*[mm]	Z**[mm]		
	first segment/upper wall					
Thermocouple	TCLU	01	1	0,5	T	±0,5 K
Thermocouple	TCLU	11	1	17,5	T	±0,5 K
Thermocouple	TCLU	12	2	21	T	±0,5 K
Thermocouple	TCLU	13	3	24,5	T	±0,5 K
Thermocouple	TCLU	21	1	34,5	T	±0,5 K
Thermocouple	TCLU	31	1	51,5	T	±0,5 K
Thermocouple	TCLU	32	2	55	T	±0,5 K
Thermocouple	TCLU	33	3	58,5	T	±0,5 K
Thermocouple	TCLU	41	1	68,5	T	±0,5 K
Thermocouple	TCLU	51	1	85,5	T	±0,5 K
Thermocouple	TCLU	52	2	89	T	±0,5 K
Thermocouple	TCLU	53	3	92,5	T	±0,5 K
Thermocouple	TCLU	61	1	102,5	T	±0,5 K
Thermocouple	TCLU	71	1	119,5	T	±0,5 K
Thermocouple	TCLU	72	2	123	T	±0,5 K
Thermocouple	TCLU	73	3	126,5	T	±0,5 K
Thermocouple	TCLU	81	1	136,5	T	±0,5 K
	First segment /down wall					
	Name		D*[mm]	Z**[mm]		
Thermocouple	TCLD	41	1	68,5	T	±0,5 K
Thermocouple	TCLD	50	0	85,5	T	±0,5 K
Thermocouple	TCLD	61	10	102,5	T	±0,5 K
Thermocouple	TCLD	70	0	119,5	T	±0,5 K
	Second segment/upper wall					
Thermocouple	TCSU	01	1	153,5	T	±0,5 K
Thermocouple	TCSU	11	1	170,5	T	±0,5 K
Thermocouple	TCSU	21	1	187,5	T	±0,5 K
Thermocouple	TCSU	31	1	204,5	T	±0,5 K
Thermocouple	TCSU	41	1	221,5	T	±0,5 K
Thermocouple	TCSU	51	1	238,5	T	±0,5 K
Thermocouple	TCSU	61	1	255,5	T	±0,5 K
Thermocouple	TCSU	71	1	272,5	T	±0,5 K

Table 7: Thermocouples Positions in the Combustion Chamber

*D= distance from the hot wall (D=0 is hot wall)

**Z= axial coordinate (Z=0 is the faceplate)

Sensor	Name	Position	Type	Precision
		Z**[mm]		
Pressure transducer	PC0	0,5	WIKA A 10	15 mbar
Pressure transducer	PC1	34,5	WIKA A 10	15 mbar
Pressure transducer	PC2	68,5	WIKA A 10	15 mbar
Pressure transducer	PC3	102,5	WIKA A 10	15 mbar
Pressure transducer	PC4	136,5	WIKA A 10	15 mbar
Pressure transducer	PC5	170,5	WIKA A 10	15 mbar
Pressure transducer	PC6	204,5	WIKA A 10	15 mbar
Pressure transducer	PC7	238,5	WIKA A 10	15 mbar
Pressure transducer	PC8	272,5	WIKA A 10	15 mbar

Table 8: Pressure Measurements Positions in the Combustion Chamber

**Z= axial coordinate (Z=0 is the faceplate)

8 Annex 3: Wall and Thermocouple Temperatures

z^{**}	T_{up}	T_{side}
[mm]	[K]	[K]
1	294,9	294,9
11	296,1	296,1
21	300,9	300,8
31	323,4	323,3
41	337,8	337,8
51	347,3	347,2
61	355,3	355,2
71	362,8	362,7
81	369,9	369,8
91	376,9	376,8
101	383,6	383,5
111	390,1	390,0
121	396,5	396,3
131	402,7	402,5
141	408,6	408,5
151	414,4	414,2
161	420,0	419,8
171	425,4	425,2
181	430,6	430,4
191	435,6	435,4
201	440,5	440,3
211	445,2	444,9
221	449,7	449,4
231	454,0	453,8
241	458,1	457,9
251	462,1	461,9
261	466,0	465,7
271	469,7	469,4
281	473,4	473,2
291	477,6	477,2
301	483,3	482,8
311	494,2	492,5
321	527,3	517,5
331	617,2	582,7

Table 9: Temperature values reconstructed from Thermtest [4].

** z = axial coordinate ($z=0$ is the start of the first segment, 29 mm before the faceplate)

Sensor name	Temperature
[-]	[K]
TCLD41	355,7
TCLD50	385,4
TCLD61	377,5
TCLD70	407,6
TCLU01	309,9
TCLU11	337,5
TCLU12	330,8
TCLU13	324,9
TCLU21	340,6
TCLU31	347,4
TCLU32	344,1
TCLU33	343,7
TCLU41	360,4
TCLU51	367,9
TCLU52	360,4
TCLU53	357,7
TCLU61	375,2
TCLU71	380,1
TCLU72	377,4
TCLU73	374,7
TCLU81	396,7
TCSU01	399,7
TCSU11	409,5
TCSU21	415,7
TCSU31	424,8
TCSU41	426,4
TCSU51	433,1
TCSU61	434,1
TCSU71	437,3
TCLU1S	299,8
TCLU5S	300,2
TCLS5S	294,8
TCSU1S	310,0
TCSU7S	314,6

Table 10: Thermocouples readings

9 References

- [1] M.P. Celano, S. Silvestri, G. Schlieben, C. Kirchberger, H. J. Haidn, T. Dawson, R. Ranjan and S. Menon. “Numerical and Experimental Investigation for a GOX-GCH4 Shear-Coaxial Injector Element”, Space Propulsion Conference 2014
- [2] S. Silvestri, M. P. Celano, G. Schlieben, C. Kirchberger, O. J. Haidn. „Characterization of a GOX-GCH4 single element Combustion chamber“,Space Propulsion Conference 2014
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